CE543: Introduction to Geotechnical Earthquake Engineering

Homework Assignment #5

March 10th, 2026

To: Dr. Arash Khosravifar
From: Brandon Auyeung

Problem 1.

A series of undrained cyclic simple shear tests were performed on K_o-consolidated specimens of saturated sand at a relative density of 50%. The cyclic loading consisted of uniform sinusoidal shear loads. Tests were performed at three different initial vertical effective consolidation stresses, with the results tabulated below.

Tests with σ_{vc}'=40 kPa			Tests with σ_{vc}'=160 kPa			Tests with σ_{vc}'=600 kPa		
Test #	Cyclic stress amplitude, τ_{cyc} (kPa)	Number of cycles to r_u=100%	Test #	Cyclic stress amplitude, τ_{cyc} (kPa)	Number of cycles to r_u=100%	Test #	Cyclic stress amplitude, τ_{cyc} (kPa)	Number of cycles to r_u=100%
1a	10.9	3	2a	41.8	2	3a	100.7	5
1b	7.2	8	2b	26.2	9	3b	71.0	13
1c	5.7	20	2c	20.0	17	3c	68.1	18
1d	4.8	28	2d	16.7	33	3d	41.0	62

Find:

Part A

Estimate the cyclic shear stress that would be required to trigger r_u=100% in 15 uniform undrained loading cycles for a specimen of this sand having an initial vertical consolidation stress of 450 kPa.

Part B

For the cyclic resistance ratio (CRR) defined as the cyclic stress ratio that triggers ru=100% in 15 uniform cycles of loading, determine the value of K_σ for an effective consolidation stress of 450 kPa.

Solution:

Part A

The CSR can be obtained by dividing the τ_{cyc} value by the effective consolidation stress that is used to consolidate the sample.

Test	CSR, $\sigma'_{3c} = 40$ kPa	CSR, $\sigma'_{3c} = 160$ kPa	CSR, $\sigma'_{3c} = 600$ kPa
a	0.272500	0.261250	0.167833
b	0.180000	0.163750	0.118333
c	0.142500	0.125000	0.113500
d	0.120000	0.104375	0.068333

Table 1. Values of CSR per test letter.

The values of CSR can be fit to a power fit with the number of cycles to failure (i.e. N to failure) as the independent values (i.e. CSR = αN^{β}).

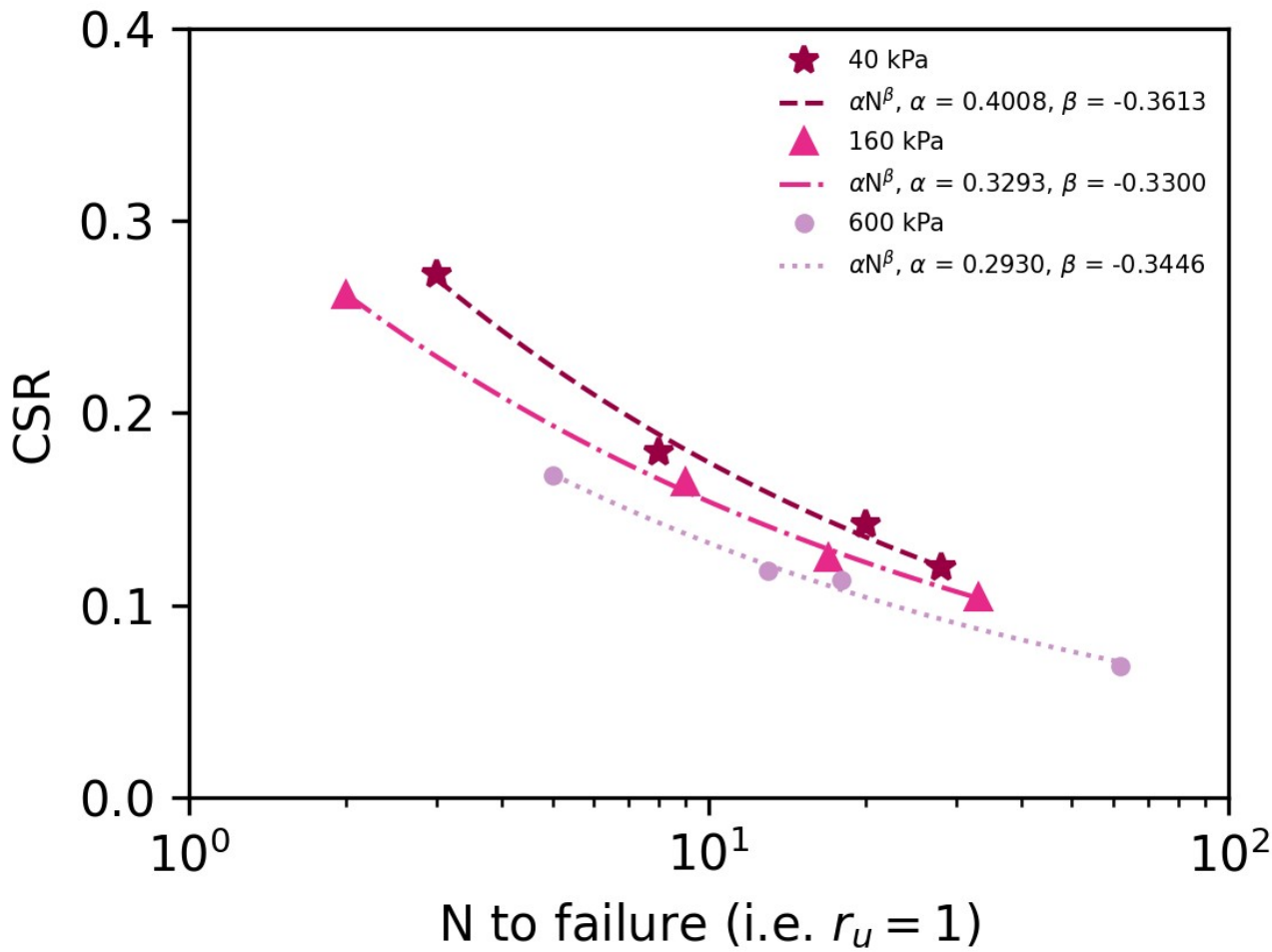

Figure 1. CSR plotted with respect to N to failure, along with power fit model on different effective consolidation stresses.

The CSR at N = 15 for σ_{3c}' = 450 kPa is between the power fits of CSR at N = 15 for σ_{3c}' = 160 kPa and CSR at N = 15 for σ_{3c}' = 600 kPa. Linear interpolation can be used to determine the value of CSR at N = 15 for σ_{3c}' = 450 kPa and thus the value of τ_{cyc} at N = 15 for σ_{3c}' = 450 kPa.

$$\frac{CSR_{600}(N=15) - CSR_{160}(N=15)}{\Delta\sigma'_{3c}} = \frac{CSR_{450}(N=15) - CSR_{160}(N=15)}{\Delta\sigma'_{3c}}$$

$$\frac{0.1347 - 0.1152}{600 \text{ kPa} - 160 \text{ kPa}} = \frac{CSR_{450}(N=15) - 0.1152}{450 \text{ kPa} - 160 \text{ kPa}}$$

Linear Interpolation

$$CSR_{450}(N=15) = 0.1219$$

$$CSR_{450}(N=15) = \frac{\tau_{cyc}}{\sigma'_{3c}} \Rightarrow \tau_{cyc} = \sigma'_{3c} CSR_{450}(N=15) = 0.1219(450 \text{ kPa})$$

$$\boxed{\tau_{cyc} = 54.8 \text{ kPa}}$$

PART B

The value of K_σ requires the CRR at 1 atm (i.e. 100 kPa). This value can be found via linear interpolation between the power fits.

$$\frac{CSR_{160}(N=15) - CSR_{40}(N=15)}{\Delta\sigma'_{3c}} = \frac{CSR_{100}(N=15) - CSR_{40}(N=15)}{\Delta\sigma'_{3c}}$$

$$\frac{0.1152 - 0.1506}{160\ \text{kPa} - 40\ \text{kPa}} = \frac{CSR_{100}(N=15) - 0.1506}{100\ \text{kPa} - 40\ \text{kPa}}$$

Linear Interpolation

$$CSR_{100}(N=15) = 0.1427$$

$$K_\sigma(\sigma'_{3c} = 450\ \text{kPa}) = \frac{CSR_{450}}{CSR_{100}} = \frac{0.1219}{0.1427}$$

$$\boxed{K_\sigma = 0.854}$$

Problem 2.

GIVEN:

The soil profile at a level site consists of a 3-m thick layer of sandy clay, overlaying an 8-m thick layer of loose sand, overlaying stiff gravelly clays that extend to depths greater than 20 m. The water table is at a depth of 2 m and the ground water is initially hydrostatic. The total unit weight of the upper sandy clay, above or below the water table, is 18 kN/m^3. The total unit weight of the loose sand layer is 20 kN/m^3. The coefficient of lateral earth pressure at rest (K_o) in the loose sand is estimated to be 0.5. Two open standpipe piezometers were installed that had small enough slotted intervals that they provide essentially point measurements of ground water conditions: Piezometer A has its slotted interval at a depth of about 4 m, while Piezometer B has its slotted interval at a depth of about 10m.

FIND:

PART A

For the initial conditions, determine: (i) where the water levels (relative to the ground surface) would be in both piezometers, and (ii) the average hydraulic gradient between the slotted intervals of the two piezometers.

PART B

For the initial conditions, determine the total horizontal stress at the depth of the slotted interval for Piezometer B.

PART C

Suppose an earthquake causes liquefaction (100% excess pore pressure ratio) in the loose sand layer. Determine: (i) where the water levels (relative to the ground surface) would be in both piezometers if they had sufficiently tall "riser" sections – i.e., portions that extend above ground, and (ii) the average hydraulic gradient between the depths of the slotted intervals for the two piezometers.

PART D

For the conditions given in (c), determine the total horizontal stress at the depth of the slotted interval for Piezometer B if the value of Ko is unaffected.

SOLUTION:

PART A

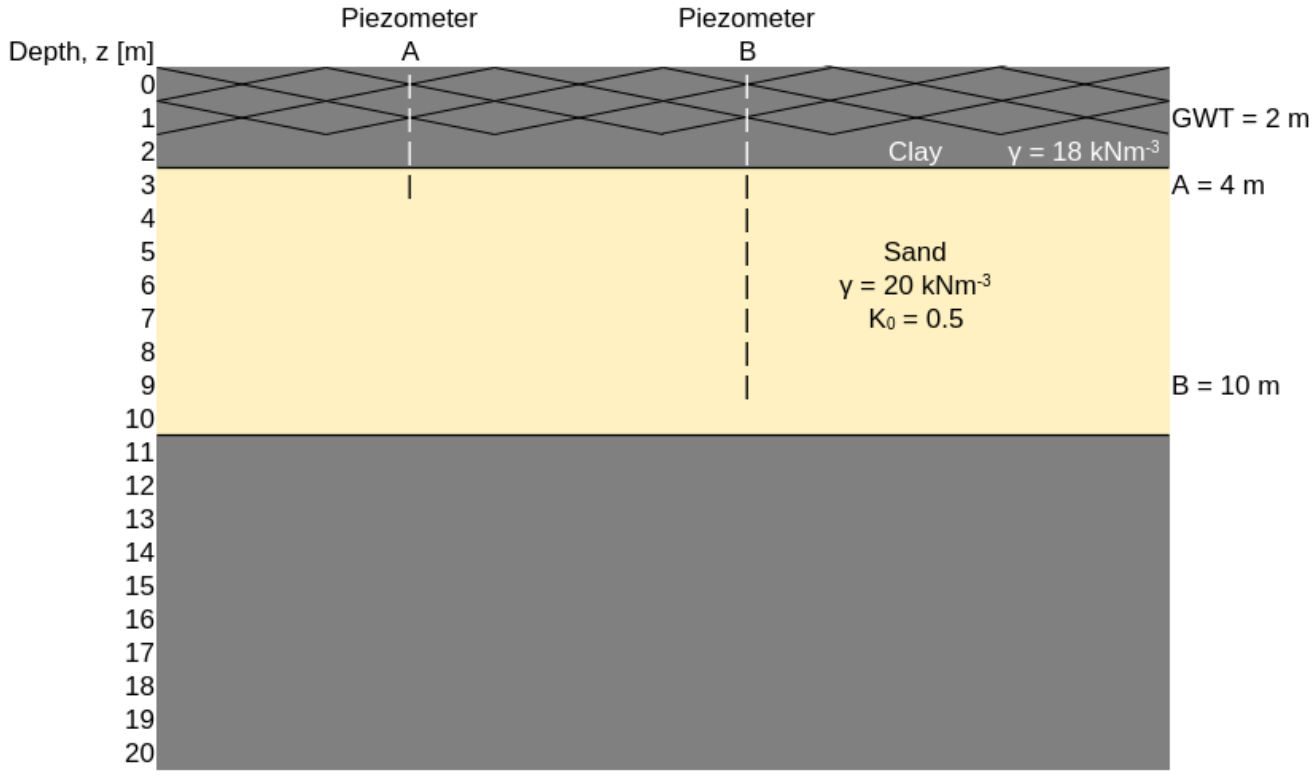

Figure 2. Rough sketch of *Problem 2*.

Piezometer	Elevation Head [m], h_z	Pressure Head [m], h_p	Total Head [m], h_T
A	6	2	8
B	0	8	8

Table 2. Hydrostatic head on piezometers using bottom of piezometer B as the datum.

The water levels in the piezometers will be where the groundwater table is (i.e. z = 2 m).

$$i_{A\to B} = \frac{\Delta h_{T,A\to B}}{\Delta L} = \frac{8\ m - 8\ m}{10\ m - 4\ m} = 0$$

PART B

$$\sigma'_{hB} = K_0\sigma'_{vB} = K_0\left[\gamma_{clay}(3\ m) + \gamma_{sand}(7\ m) - \gamma_w(8\ m)\right]$$

$$\sigma'_{hB} = 0.5\left[18\ \text{kNm}^{-3}(3\ m) + 20\ \text{kNm}^{-3}(7\ m) - 9.81\ \text{kNm}^{-3}(8\ m)\right] = 57.8\ \text{kPa}$$

$$\sigma_{hB} = \sigma'_{hB} + u_B = 57.8\ \text{kPa} + \gamma_w(8\ m) = 57.8\ \text{kPa} + 9.81\ \text{kNm}^{-3}(8\ m)$$

$$\boxed{\sigma_{hB} = 136.8\ \text{kPa}}$$

PART C

Liquefaction can only occur in the loose sand layer. When liquefaction occurs, the pressure ratio, r_u, is equal to one (i.e. $\Delta u = \sigma_{v0}'$).

z [m]	σ_v [kPa]	u_0 [kPa]	σ'_{v0} [kPa]	Δu [kPa]	u_f [kPa]	σ'_{vf} [kPa]
0	0	0.000000	0.000000	0.000000	0.000000	0.000000
1	18	0.000000	18.000000	0.000000	0.000000	18.000000
2	36	0.000000	36.000000	0.000000	0.000000	36.000000
3 (End of Clay)	54	9.810000	44.190000	0.000000	9.810000	44.190000
3 (Start of Sand)	54	9.810000	44.190000	44.190000	54.000000	0.000000
4	74	19.620000	54.380000	54.380000	74.000000	0.000000
5	94	29.430000	64.570000	64.570000	94.000000	0.000000
6	114	39.240000	74.760000	74.760000	114.000000	0.000000
7	134	49.050000	84.950000	84.950000	134.000000	0.000000
8	154	58.860000	95.140000	95.140000	154.000000	0.000000
9	174	68.670000	105.330000	105.330000	174.000000	0.000000
10	194	78.480000	115.520000	115.520000	194.000000	0.000000
11	214	88.290000	125.710000	125.710000	214.000000	0.000000

Table 3. Values of stresses and pore water pressures at depths.

The hydraulic gradient and be calculated by obtaining the total heads after liquefaction at the two piezometers. The datum of ground surface will be used here.

$$h_{T,A} = z_A + \frac{u_{f,A}}{\gamma_w} = -4\ \text{m} + \frac{74\ \text{kPa}}{9.81\ \text{kNm}^{-3}} = 3.543\ \text{m}$$

$$h_{T,B} = z_B + \frac{u_{f,B}}{\gamma_w} = -10\ \text{m} + \frac{194\ \text{kPa}}{9.81\ \text{kNm}^{-3}} = 9.776\ \text{m}$$

$$i_{B\to A} = \frac{\Delta h_{T,B\to A}}{\Delta L} = \frac{9.776\ m - 3.543\ m}{10\ m - 4\ m} = 1.04$$

Liquefaction has occurred.

Water is expected to rise to the surface of the ground within the piezometers (i.e. z = 0 m).

PART D

$\sigma'_{hB,f} = 0$ *No effective stress in liquefied layer.*

$$\sigma_{hB,f} = \sigma'_{hB,f} + u_{B,f} = 0 + 194 \text{ kPa} = 194 \text{ kPa}$$

Problem 3.

GIVEN:

Square footings of width B will support columns for an elevated bridge that crosses over a The liquefaction resistance of a clean, poorly graded sand (SP) deposit was evaluated using cyclic triaxial testing of undisturbed samples obtained using frozen sampling techniques. The sand deposit extends from the ground surface to a depth of 10 m. The water table is at a depth of 2 m, and ground water conditions are initially hydrostatic. The sand is normally consolidated with an estimated coefficient of earth pressure at rest of 0.45. Isotropically consolidated undrained cyclic triaxial tests were performed on samples from a depth of 5 m with the results presented below.

Tests with $_{3c}$'=70 kPa		
Test #	Cyclic stress amplitude, $q_{cyc}/2$ (kPa)	Number of cycles to r_u=100%
1	38.6	2.25
2	32.3	5
3	21.0	14
4	20.2	21
5	15.4	36

FIND:

PART A

For the initial conditions, estimate the vertical effective stress at a depth of 5 m.

PART B

Suppose the site was shaken by an earthquake, with each direction of horizontal shaking producing cyclic shear stresses at a depth of 5 m that are equivalent to about 10 uniform cycles at a cyclic stress ratio of 0.14. Assuming the sand is essentially undrained during shaking, and using the mid-range of the r_u versus FS_{liq} relation by Marcuson and Hynes (1989) (Figure 31 below from the textbook), estimate the vertical effective stress at that depth.

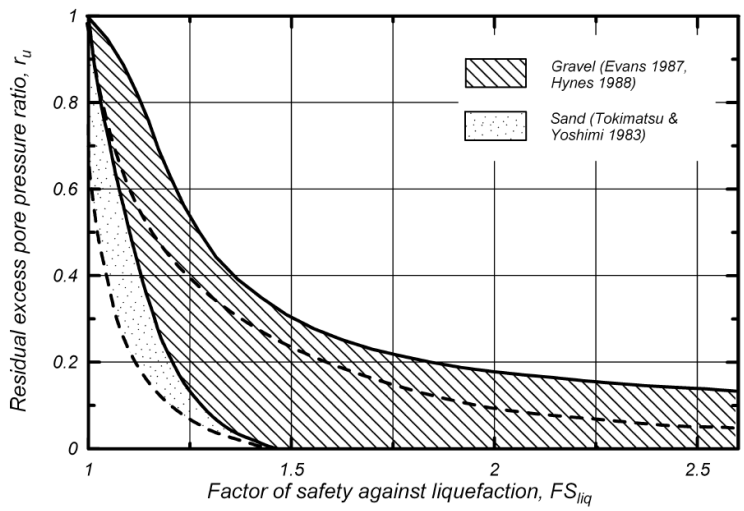

Figure 31. Excess pore water pressure ratio versus FS_{liq} under level ground conditions, from laboratory test data (after Marcuson et al. 1990).

PART C

Repeat part (b) for an earthquake that causes about the same number of loading cycles but with amplitudes that are about twice as large.

SOLUTION:

PART A

We assume sand to have a unit weight of 20 kNm^{-3}.

$$\sigma'_{v0} = \sigma_{v0} - u = \gamma z - \gamma_w(z - z_w) = 20\ \mathrm{kNm^{-3}}(5\ \mathrm{m}) - 9.81\ \mathrm{kNm^{-3}}(5\ \mathrm{m} - 2\ \mathrm{m}) = 70.6\ \mathrm{kPa}$$

PART B

The CSR of the soil can be obtained by dividing the τ_{cyc} values (which are equal to $0.5q_{cyc}$) by σ_{vc}' used to consolidate, this value being 70 kPa.

The CSR values obtained can be translated into in-situ field shaking with the following equation.

$$CSR_{field} = 0.9\frac{1+2K_0}{3}CSR$$

N to failure	τ_{cyc} [kPa]	CSR	CSR_{field}
2.25	38.6	0.551429	0.314314
5	32.3	0.461429	0.263014
14	21.0	0.300000	0.171000
21	20.2	0.288571	0.164486
36	15.4	0.220000	0.125400

Table 4. CSR values.

Power fits can be obtained on both CSR values and graphed as seen below.

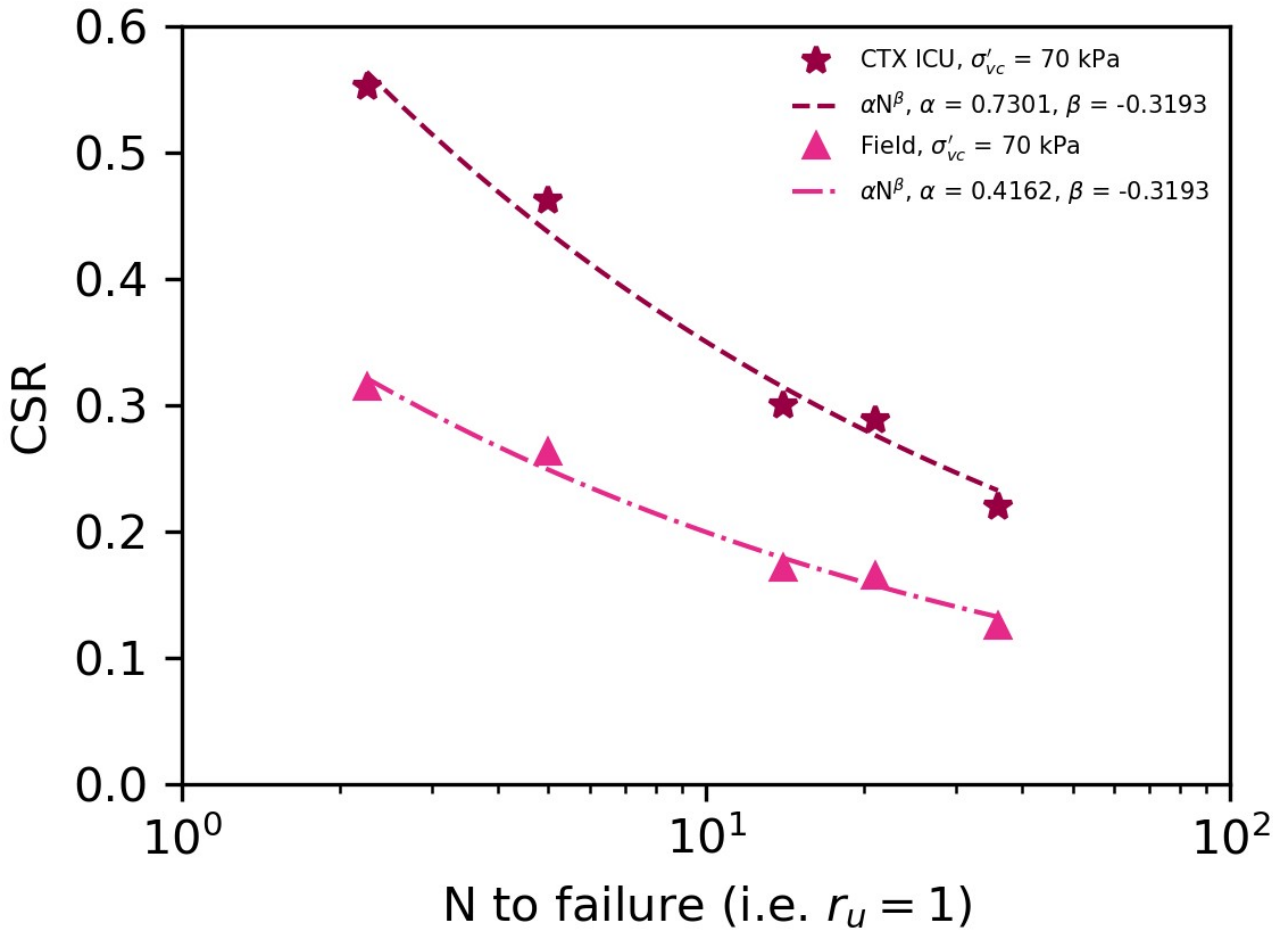

Figure 3. CSR values and power fits for CTX ICU and field shaking.

The CSR of 0.14 due to an EQ corresponds to ten uniform cycles (i.e. N = 10). Therefore, the CRR is the soil's in-situ capacity at N = 10.

$$CRR = CSR_{field}(N = 10) = 0.4161(10)^{-0.3193} = 0.20$$

$$FS_{liq} = \frac{CRR}{CSR} = \frac{0.20}{0.14} = 1.42$$

Via Marcuson and Hynes (1989), a r_u value of 0.14 can be obtained. The r_u can be used to obtain the values necessary to calculate the vertical effective stress during shaking.

$$r_u = \frac{\Delta u}{\sigma'_{vc}} \Rightarrow \Delta u = r_u \sigma'_{vc} = 0.14(70 \text{ kPa}) = 9.8 \text{ kPa}$$

$$\boxed{\sigma'_{vf}(N = 10) = \sigma'_{v0} - \Delta u = 70.6 \text{ kPa} - 9.8 \text{ kPa} = 60.9 \text{ kPa}}$$

Part C

Because the amplitude has doubled with no change in number of cycles, the CSR can also be doubled. The new CSR is equal to twice of 0.14 (i.e. 0.28). The CSR is now greater than the CRR. This means the soil has liquefied, resulting in an r_u equal to 1.

Therefore, the vertical effective stress after doubling the EQ amplitude is 0, since the soil is liquefied.